TEXORA · LECTURE SHEET

Permeability, Crease Recovery, Pilling and Flame Tests

Beyond strength, a fabric must breathe, resist water when required, spring back from creasing, stay smooth in wear and behave safely near flame. This sheet covers air and water-vapour permeability, hydrostatic head, crease recovery, pilling and flammability — the comfort, appearance and safety tests of fabric evaluation.

TE-3201 · QUALITY EVALUATION OF FABRIC AND APPAREL · TOPIC 6 · LEVEL 3 TERM 2

1. Why these properties matter

Strength tests tell whether a fabric will survive. The tests in this sheet tell whether it will be *acceptable*: comfortable to wear, presentable after a day of use, durable in appearance and safe in its end use. They are sometimes grouped as comfort properties (permeability), aesthetic or appearance properties (crease recovery, pilling) and safety properties (flammability). All are measured on fabric conditioned in the standard atmosphere — 65 % RH, 20 °C — because moisture content changes pore size, fibre stiffness and burning behaviour. A result without its test conditions is not comparable with any other laboratory's result.

Two cautions run through the whole sheet. First, most of these tests are *comparative*: the instrument applies a standardised insult and the fabric is scored against a standard, a scale or a specification, not against a universal physical constant. Second, fabric direction matters. Warp and weft behave differently in creasing and permeability, so specimens are cut and reported in both directions.

2. Air permeability — ISO 9237

Air permeability is the rate at which air passes through a fabric under a defined pressure difference. It governs wind resistance, thermal comfort, drying rate and the performance of parachutes, airbags, filters and industrial fabrics. A shirting that passes no air feels clammy; a tent fabric that passes too much air will not hold warmth; a parachute canopy must pass a tightly specified amount — too little and descent is unstable, too much and descent is too fast.

2.1 Principle and apparatus

In ISO 9237 the specimen is clamped over a circular test area — commonly 20 cm^2 — and air is drawn through it so that a *defined pressure drop* is maintained across the fabric. The standard pressure drop

depends on end use: 100 Pa for apparel fabrics, 200 Pa for industrial fabrics, with other values by agreement. The volume flow rate at that pressure drop is measured, historically by a variable-area flowmeter or by pressure drop across a calibrated orifice, and in modern testers by electronic flow sensing. The result is expressed as velocity of air through unit area:

$$\text{Air permeability} = \text{volume flow rate} / \text{test area} \quad [\ \text{cm}^3/\text{cm}^2/\text{s}\ ]$$

The unit $\text{cm}^3/\text{cm}^2/\text{s}$ is identical in meaning to mm/s, and many reports give mm/s or $\text{L/m}^2/\text{s}$ (1 $\text{cm}^3/\text{cm}^2/\text{s}$ = 10 $\text{L/m}^2/\text{s}$). The clamping ring must seal the edge without stretching or crushing the fabric; leakage around the edge is the classic source of a falsely high result, so a guard ring and a leak check with an impermeable blank are part of good practice. At least five, preferably ten, specimens are tested at different places on the sample, avoiding selvedges, creases and faults, and the mean and coefficient of variation are reported together with test area, pressure drop and atmosphere.

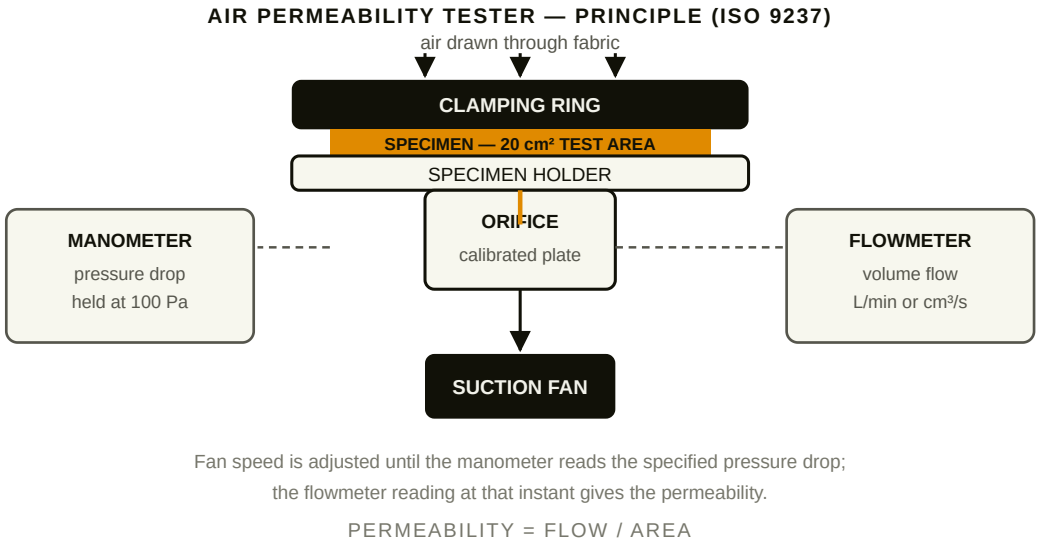


Figure 1 — Air permeability apparatus. The specimen is sealed in the holder, suction holds a defined pressure drop across it, and the measured flow divided by test area gives the result.

WORKED EXAMPLE — AIR PERMEABILITY

A cotton poplin is tested at 100 Pa on a 20 cm^2 area. The flowmeter reads 2.8 L/min.

Convert the flow: 2.8 L/min = 2,800 cm^3 per 60 s = 2,800 / 60 = **46.7 cm^3/s**.

Permeability = 46.7 / 20 = **2.33 $\text{cm}^3/\text{cm}^2/\text{s}$** (equivalently 2.33 mm/s, or 23.3 $\text{L/m}^2/\text{s}$).

This is a fairly closed apparel fabric — suitable where wind resistance is wanted. An open summer voile might read ten times higher; a coated parachute fabric is specified far lower. The number is meaningless unless the 100 Pa pressure drop and 20 cm^2 area are reported with it.

2.2 What controls air permeability?

- **Cover and porosity** — the dominant factor. Tighter weave, higher thread density and thicker yarns close the inter-yarn pores and permeability falls steeply.
- **Yarn structure** — hairy, bulky yarns block pores more than smooth filament yarns at the same cover; twist compacts the yarn and opens the pores slightly.
- **Finishing** — calendering flattens yarns and closes pores; resin and coating treatments can reduce permeability almost to zero; raising and brushing block pores with surface fibre.
- **Moisture** — hygroscopic fibres swell when damp, closing pores, so a cotton fabric is less permeable wet than dry.

3. Water-vapour permeability and resistance to water penetration

3.1 Water-vapour permeability (breathability)

A garment can block liquid water yet still feel comfortable if water *vapour* from perspiration escapes through it. Water-vapour permeability is measured by cup methods: the specimen seals a dish of water (or of desiccant) and the dish is weighed after a period in a controlled atmosphere; mass loss per unit area per unit time (g/m^2/24 h) is the vapour transmission rate. Dish method variants differ in whether the water touches the fabric or an air gap is left, and results from different variants are not interchangeable. Microporous and hydrophilic coatings achieve high vapour transmission with zero liquid penetration — the basis of modern waterproof-breathable rainwear. Report the method, temperature and humidity gradient with every value.

3.2 Hydrostatic head — ISO 811

Resistance to liquid-water penetration is measured as **hydrostatic head**: the specimen is clamped and water pressure on one face is raised at a standard rate until penetration occurs at the third separate place on the specimen. The pressure at that point, in cm of water column (or Pa), is the result. A groundsheet or tent fly needs a high head; a showerproof finish gives a modest head that fails under sustained pressure; untreated cotton gives almost none. Seams, needle holes and coatings cracks — not the intact fabric — are usually where a garment first leaks, which is why hydrostatic head on fabric must be read together with garment testing (Topic 10).

PROPERTY	WHAT PASSES	DRIVING FORCE	TYPICAL RESULT UNIT
Air permeability	Air (bulk flow)	Pressure drop, e.g. 100 Pa	$cm^3/cm^2/s$
Water-vapour permeability	Water vapour (diffusion)	Vapour-pressure gradient	g/m^2/24 h

Hydrostatic head	Nothing — pressure resisted	Rising water pressure	cm H_2O or Pa

4. Crease recovery — ISO 2313

Creasing is bending taken past recovery: fibres on the outside of the fold are stretched, those inside compressed, and if the deformation is held, molecular bonds re-form in the creased shape. Crease recovery measures how completely the fabric springs back when the creasing load is removed. It predicts the fresh or tired look of a garment after sitting, packing and wearing.

4.1 The angle method

In ISO 2313 a strip of fabric is folded in half and loaded with a standard weight for a standard time (commonly 1 kg for 5 minutes), then released and allowed to recover for a standard time (commonly 5 minutes) with one limb held horizontal and the other hanging free. The angle between the two limbs — the **crease recovery angle** — is measured. A perfect recovery gives 180°; a fabric that stays folded gives near 0°. Warp-way and weft-way specimens are tested separately, face-to-face and back-to-back foldings are distinguished in careful work, and results are often summed:

$$\text{Total crease recovery} = \text{warp recovery angle} + \text{weft recovery angle} \quad (\text{maximum } 360°)$$

WORKED EXAMPLE — READING CREASE RECOVERY

A polyester–cotton shirting gives warp recovery 127° and weft recovery 115°.
Total = 127 + 115 = **242° of a possible 360°**, i.e. about 67 % angular recovery.
Reading: respectable easy-care behaviour for a blend, short of a fully resin-finished cotton (which can exceed 280°) and well above an untreated cotton (often 150–180° total). The warp–weft gap (12°) is normal — warp yarns are usually more highly tensioned and finished — but a very large gap warns of uneven finishing.

4.2 What improves recovery?

Resilient fibres recover best: wool and polyester spring back because of their molecular structure, while cotton and viscose, whose hydrogen bonds re-form in the bent position, crease badly unless crosslinked by an easy-care resin finish. Yarn twist, fabric thickness and weave all modify the result — thicker fabrics and looser weaves generally recover better because individual yarns bend less sharply. Note the trade-off: the resin finishes that raise crease recovery also reduce tear strength and abrasion resistance, so the specification must balance appearance against durability.

5. Pilling — ISO 12945

5.1 How a pill forms

Pilling proceeds in stages. (1) **Fuzzing**: rubbing works loose fibre ends out of the yarn to form a surface fuzz. (2) **Entanglement**: continued rubbing rolls the fuzz into small balls — pills — anchored to the surface by a few strong fibres. (3) **Wear-off**: pills eventually break away if their anchor fibres are weak. Whether a fabric looks pilled therefore depends on the balance between pill formation and pill removal. This explains a classic paradox: strong synthetic fibres such as polyester and nylon form pills that *persist*, because the anchor fibres do not break, whereas weak fibres such as cotton form pills that fall off — a cotton fabric may pill less visibly than a stronger polyester fabric of identical construction. Blends of strong synthetics with weak cellulosics are the worst case: the cellulosic fuzz is anchored by indestructible synthetic fibres.

5.2 Test methods and grading

ISO 12945 covers two laboratory simulations. In the **Martindale method** specimens rub against a standard abradant (often the same fabric) in a Lissajous motion under light pressure for a set number of rubs. In the **ICI pilling box method** specimens mounted on rubber tubes tumble inside a cork-lined box for a set number of revolutions. After the prescribed rubbing, trained assessors grade the surface under standard lighting against written descriptions or photographic standards:

GRADE	DESCRIPTION
5	No change — surface as original
4	Slight surface fuzzing and/or partially formed pills
3	Moderate fuzzing and/or pilling — pills of varied size cover the surface partially
2	Distinct pilling — large pills cover a large part of the surface
1	Severe pilling — dense pills cover the whole surface

Half-grades (e.g. 3–4) are permitted. Grading is done at staged intervals — a fabric may be grade 4–5 at 2,000 rubs and grade 3 at 7,000 rubs — and the full series is reported, not only the final grade. Anti-pill routes include using longer, coarser or lower-persistence fibres, higher yarn twist, singeing and shearing, compact constructions and anti-pill finishes; every route is verified by re-testing, because a change that reduces fuzzing may not reduce persistent pills.

6. Flammability

6.1 Concepts: flammable, flame-retardant, flame-resistant

Three terms must not be confused. A **flammable** fabric ignites readily and burns rapidly. A **flame-retardant** fabric has been treated (or built) so that it ignites with difficulty, burns slowly or self-extinguishes when the flame is removed — the retardancy may be a durable finish, a back-coating or a topical treatment that washes out. A **flame-resistant** (inherently flame-retardant) fabric resists burning by its fibre chemistry itself — aramids, modacrylics, flame-resistant viscose — and the property survives laundering because it is in the polymer. Cotton treated with a durable phosphorus finish is flame-retardant; an aramid workwear fabric is flame-resistant; untreated cotton is neither.

6.2 Limiting oxygen index (LOI)

The **limiting oxygen index** is the minimum oxygen concentration, in a flowing oxygen–nitrogen mixture, that just supports candle-like burning of a small specimen. Air contains about 21 % oxygen, so a material with LOI below 21 burns freely in air, one near 21 is marginal, and one well above 21 tends to self-extinguish in air. Typical fibre values: **cotton ≈ 18 %** (burns readily), acrylic ≈ 18 %, polyester ≈ 20–21 %, **wool ≈ 25 %** (reluctant to ignite, chars, self-extinguishes — the reason wool is favoured for aircraft and contract interiors), and aramids 28–30 % and above. LOI is a small-scale material index, not a garment safety certificate: fabric weight, construction, blends and finishes shift real behaviour, and thermoplastic fibres complicate the picture by melting and dripping away from the flame — which can either remove fuel or spread burning drops.

6.3 Vertical-strip and related burning tests

In the **vertical-strip test** a strip is suspended vertically, a standard small flame is applied to its lower edge for a set time, and the laboratory records afterflame time (how long it continues to flame), afterglow time, char length (the damaged length under a tearing load) and whether flaming debris or melt-drips fall. A fabric that self-extinguishes with a short char length passes the typical protective-clothing criterion; one whose flame spreads to the top of the strip fails outright. Children's nightwear, upholstery and protective clothing each have their own regulated test geometries — 45° incline and horizontal tests also exist — so the report must name the exact method. Laboratory burning tests use small, controlled flames; they rank materials, but no laboratory pass guarantees behaviour in a real fire, where heat flux, ventilation and garment fit dominate.

FIBRE	APPROX. LOI (%)	BEHAVIOUR IN AIR
Cotton	18	Ignites readily, burns steadily, afterglow
Polyester	20–21	Melts, shrinks from flame, burns slowly with dripping
Wool	25	Difficult to ignite, chars, usually self-extinguishes
Aramid	28–30	Does not melt, chars, flame-resistant by chemistry

Glossary

Air permeability — volume flow of air per unit area at a defined pressure drop. **Hydrostatic head** — water pressure resisted before penetration at a third place. **Water-vapour permeability** — rate of moisture-vapour transmission through a fabric. **Crease recovery angle** — angle to which a folded, loaded strip reopens after standard recovery. **Pilling** — formation of fibre balls on a fabric surface during rubbing. **LOI** — limiting oxygen index: minimum % oxygen supporting combustion. **Flame-retardant** — treated or built to retard burning. **Flame-resistant** — inherently resistant by fibre chemistry. **Char length** — damaged length of a burned specimen under a standard tearing load.

Key points

• Air permeability (ISO 9237) = flow / area at a stated pressure drop — always report area and pressure drop with the value.
• Liquid resistance (hydrostatic head, ISO 811), vapour transmission and air flow are three different properties; a fabric can excel at one and fail at another.
• Crease recovery (ISO 2313) is an angle; warp + weft totals are compared against 360°.
• Pilling is graded 5 (none) to 1 (severe); strong fibres make pills persist, so strength can worsen appearance.
• LOI ranks burning tendency — cotton ≈ 18 %, wool ≈ 25 % — but small-scale tests rank materials, they do not certify real-fire safety.

Practice

1. A fabric passes 3.6 L/min through a 20 cm^2 area at 100 Pa. Calculate its air permeability in $cm^3/cm^2/s$ and in $L/m^2/s$.
2. Crease recovery angles: warp 132°, weft 121°. Calculate the total recovery and comment on it for an easy-care shirting.
3. Explain why a polyester-rich blend often shows worse pilling grades than an all-cotton fabric of the same construction, even though polyester is the stronger fibre.
4. A buyer specifies LOI alone for protective trousers. Give two reasons why LOI by itself is an insufficient specification.

Answer key

1. Flow = 3,600 cm^3 / 60 s = 60 cm^3/s. Permeability = 60 / 20 = **3.0 $cm^3/cm^2/s$**. In $L/m^2/s$: 3.0 × 10 = **30 $L/m^2/s$**.

2. Total = 132 + 121 = **253° of 360°** (about 70 % angular recovery). Good easy-care performance — above a typical blend value and approaching resin-finished cotton; acceptable for a quality shirting specification.
3. Pills are anchored by fibres that survive rubbing. Polyester anchor fibres are strong and do not break, so pills formed from surface fuzz persist and accumulate. In all-cotton fabric the anchor fibres are weak, pills break off as fast as they form, and the surface stays cleaner. The blend is worst when weak fibres supply the fuzz and polyester supplies unbreakable anchors.
4. First, LOI is a fibre/material index on a tiny specimen — fabric weight, construction, blend ratio and finish change real burning behaviour. Second, LOI says nothing about melt-dripping, afterglow, char length, heat transfer to skin or behaviour after laundering, all of which the end-use regulation's strip test and garment requirements address. LOI may support screening; the specification must name the actual burning test and pass criteria.
